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Ab initio kinetics for the decomposition of monomethylhydrazine (CH_3NHNH_2)

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Abstract

The decomposition kinetics of CH_3NHNH_2 (monomethylhydrazine) is studied with *ab initio* transition state theory-based master equation analyses. The simple NN and CN bond fissions to produce the radicals $CH_3NH + NH_2$ or $CH_3 + NHNH_2$ are expected to dominate the decomposition kinetics. The transition states for these two bond fissions are studied with variable reaction coordinate transition state theory employing directly determined CASPT2/aug-cc-pVDZ interaction energies. Orientation independent corrections for limitations in the basis set and for the effects of conserved mode geometry relaxation are included. The bond dissociation energies are evaluated at the QCISD(T)/CBS//B3LYP/6-311++G(d,p) level. The transition state theory analysis directly provides high pressure dissociation and recombination rate coefficients. Predictions for the pressure dependence and product branching in the dissociation of CH_3NHNH_2 are obtained by solving the master equation.
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Keywords: Monomethylhydrazine; *Ab initio*; Transition state theory; Master equation

1. Introduction

Monomethylhydrazine (MMH) is a commonly used bipropellant and monopropellant in rocket engines [1,2]. Since MMH is prone to exothermically decompose upon contact with either a hot surface or an oxidizer, the thermal decomposition of MMH is an important issue related to fuel stability and storability. Furthermore, MMH decomposition reactions are a necessary component of the detailed kinetic mechanisms for MMH oxidization employed in the numerical simulation of rocket engine combustion.

Earlier experimental investigations on the thermal decomposition of MMH include those conducted with the toluene-carrier flow technique at 0.01–0.04 atm [3], adiabatic flow reactors at atmospheric pressure and 750–1000 K [4], and low-pressure flow reactors at 850–1100 K [5]. More recently, Catoire et al. [6] performed shock tube experiments for the thermal decomposition of MMH at 1040–1370 K and 140–455 kPa, and proposed a detailed kinetic mechanism to model the experimental data.

The thermal decomposition of MMH was studied theoretically by Sun and Law [7] using *ab initio* and density functional electronic structure calculations coupled with quantum Rice–Ramsperger–Kassel (QRRK) theory and a master equation analysis for the pressure falloff. An

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elementary reaction mechanism was then developed to model the overall MMH thermal decomposition rates of Eberstein and Glassman [4] at 750–1500 K and 1 atm. The NN and CN bond fission reactions,

$$\mathrm{MMH} \rightarrow \mathrm{NH_2} + \mathrm{CH_3NH} \tag{R1}$$

and

$$\mathrm{MMH} \rightarrow \mathrm{CH_3} + \mathrm{NHNH_2} \tag{R2}$$

were found to be the dominant channels for MMH thermal decomposition. Recently, Sun et al. [8] further developed their kinetic mechanism and used it to model shock tube experimental data for the thermal decomposition of MMH at temperatures of 900–1370 K and pressures of 140–450 kPa. At elevated pressures, the MMH decomposition lifetime was found to be most sensitive to the rate coefficient for the N–N bond fission reaction, R1

The reverse of reactions R1 and R2 are the barrierless radical–radical recombination reactions

$$\mathrm{NH_2} + \mathrm{CH_3NH} \rightarrow \mathrm{MMH} \tag{R-1}$$

and

$$\mathrm{CH_3} + \mathrm{NHNH_2} \rightarrow \mathrm{MMH} \tag{R-2}$$

The application of transition state theory (TST) to such barrierless reactions, and correspondingly to the reverse simple bond fissions, is complicated by the absence of a pronounced saddle point on the potential energy surface (PES). For such reactions, there is a considerable variability to the location of the transition state on the PES. In the MMH thermal decomposition mechanism of Sun and coworkers [7,8] the rate coefficients of R1 and R2 were estimated from values for related generic reactions.

In the present study, we predict the capture rate coefficients for R1 and R2, and their reverses R-1 and R-2, with variable reaction coordinate transition state theory (VRC-TST). This approach has been shown to accurately treat the kinetics of barrierless reactions when an accurate potential energy surface describing the interaction between the radicals is available [9–11]. Its direct implementation, in which the interaction potential of the reacting fragments is evaluated on-the-fly from multireference electronic structure calculations, allows for the efficient computation of quantitatively accurate rate coefficients. In recent studies, the direct VRC-TST method was shown to provide a computationally practical means for predicting the rate coefficients for a series of hydrogen atom – hydrocarbon radical and hydrocarbon radical – hydrocarbon radical reactions [12–14]. Most recently, a theoretical framework based on the VRC-TST method was developed for studying the branching between roaming and bond fission channels in molecular dissociations and applied to the decomposition of acetaldehyde [15].

We also model the pressure dependence and product branching in R1 and R2 with a master equation (ME) analysis incorporating the transition state information obtained from the VRC-TST calculations.

2. Theoretical methods

2.1. Potential energy surface

In VRC-TST calculations the intermolecular degrees of freedom, denoted by R describing the separation of the fragments and by Ω describing the relative orientation, are treated as fully coupled anharmonic modes via classical phase space integrals. This analysis requires an interaction potential for these degrees of freedom, which was determined here with on-the-fly CASPT2 [16] calculations employing the cc-pVDZ and aug-cc-pVDZ basis sets of Dunning [17,18]. For both reactions R1 and R2 the active space for the CASPT2 calculations was chosen to be the minimum required to correctly describe the separated fragments: two electrons in two orbitals (2e,2o). These calculations of the interaction potential are denoted as $V_{\mathrm{CASPT2}/DZ}(R,\Omega;\mathrm{rigid})$ and $V_{\mathrm{CASPT2}/ADZ}(R,\Omega;\mathrm{rigid})$, respectively, where "rigid" is used to indicate that the reacting fragments have fixed internal geometries. Here, these rigid structures are obtained from B3LYP/6-311++G(d,p) [19,20] calculations of the equilibrium geometries of the isolated reactants.

Two correction terms are included in the final potential energy: an R-dependent one-dimensional correction term from CASPT2/aug-cc-pVTZ electronic structure calculations, and a one-dimensional correction for the effect of relaxing the internal structure of the reacting fragments along the minimum energy path (MEP). For the aug-cc-pVDZ case, the final potential can be expressed as,

$$\begin{aligned} V = {} & V_{\mathrm{CASPT2}/ADZ}(R,\Omega;\mathrm{rigid}) \\ & + [V_{\mathrm{CASPT2}/ATZ}(R,\Omega^*;\mathrm{rigid}) \\ & - V_{\mathrm{CASPT2}/ADZ}(R,\Omega^*;\mathrm{rigid})] \\ & + [V_{\mathrm{CASPT2}/ADZ}(R,\Omega_e;\mathrm{relaxed}) \\ & - V_{\mathrm{CASPT2}/ADZ}(R,\Omega_e;\mathrm{rigid})], \end{aligned} \tag{3}$$

where the orientation Ω^* is the equilibrium value at $R = 5.7$ au, and the orientation Ω_e is the R-dependent equilibrium value along the MEP. The expression in Eq. (3) assumes that both corrections are independent of orientation.

The CASPT2/aug-cc-pVDZ relaxed potentials along the MEPs are illustrated in Fig. 1. The $\mathrm{NH_2} + \mathrm{CH_3NH}$ potential is seen to be more

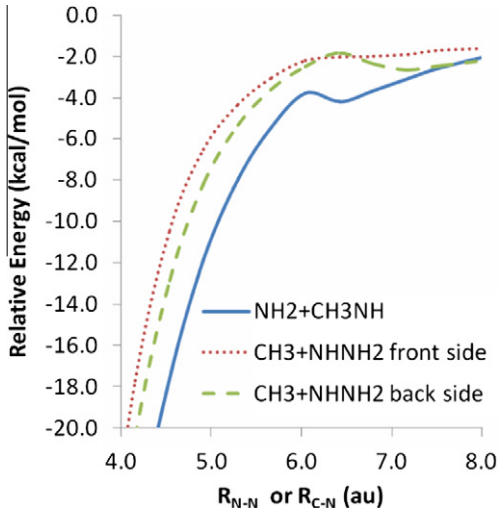

Fig. 1. Potential curves for $NH_2 + CH_3NH$ (solid line), and the front (dotted line) and back (dashed line) sides of $CH_3 + NHNH_2$.

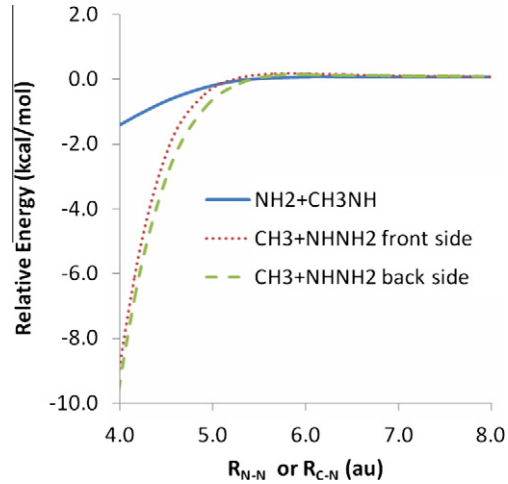

Fig. 2. Total potential correction curves for $NH_2 + CH_3NH$ (solid line), and the front (dotted line) and back (dashed line) sides of $CH_3 + NHNH_2$.

attractive than either of the $CH_3 + NHNH_2$ potentials, suggesting that the recombination rate for R-1 will be greater than that for R-2. The decreased attractiveness for the $CH_3 + NHNH_2$ case is related to the need to break the (3e,2o) NN π bond in $NHNH_2$ during the formation of the CN bond. For the R-2 reaction, there are two distinct addition sites on the $NHNH_2$ radical corresponding to the addition of the CH_3 radical to its front and back sides. The front and back sides are determined by whether the two out-of-plane N–H bonds point away from or towards the direction of the CH_3 approach. For the R-1 reaction, there are also front and back sides for the CH_3 radical, but they are symmetrically related.

Interestingly, all the potentials have a submerged barrier, which is related to the need to reorient from a long-range van der Waals orientation to a short-range chemical bonding orientation. Such reorientational minima are generally to be expected, particularly when hydrogen-bonding occurs. This reorientational saddle point and hydrogen-bonding motivated the present use of the aug-cc-pVDZ basis set, which was shown to provide an effective treatment of such interactions in our recent study of the $CH_3 + OH$ reaction [21].

The total basis set and relaxation correction is illustrated in Fig. 2 for the aug-cc-pVDZ cases. For the $NH_2 + CH_3NH$ case, the correction is only a small fraction of the total interaction potential at any given separation. In contrast, for the $CH_3 + NHNH_2$ cases, the corrections, which are primarily a relaxation effect, are a significant fraction of the total interaction at shorter separations. The relatively large magnitude of the latter corrections is likely due to the transformation from sp^2 to sp^3 hybridization in the CH_3 fragment as the CN bond is formed.

The bond dissociation energies are obtained here from quadratic configuration interaction with singles doubles and perturbative inclusion of triples (QCISD(T)) calculations. The geometries and zero-point corrections for these calculations are obtained with B3LYP/6-311++G(d,p) density functional theory. The QCISD(T) calculations are performed for the cc-pVTZ and cc-pVQZ basis sets and then extrapolated to the complete basis set limit (CBS) [22]. This QCISD(T)/CBS//B3LYP/6-311++G(d,p) method is also employed in the analysis of the torsional minima and barriers for MMH and for CH_3NH.

The final zero-point corrected QCISD(T)/CBS//B3LYP/6-311++G(d,p) dissociation energies are 62.7 and 64.6 kcal/mol, for reactions R1 and R2, respectively. Related estimates were provided in the recent study of Sun and Law [7]. This study also suggested that the tight transition state for formation of $CH_3N + NH_3$ lies slightly below this dissociation threshold. For completeness we have reexamined this TS (TS1). At the zero-point corrected QCISD(T)/CBS//B3LYP/6-311++G(d,p) level we find a barrier height of 62.9 kcal/mol, or 0.2 kcal/mol higher than the bond fission R1. This channel is not considered further here because the lower entropy and higher energy for this tight TS implies that its contribution to the kinetics will be dominated by that from R1. These QCISD(T)/CBS estimates are expected to have uncertainties of about 1 kcal/mol.

In the present study, all of the density functional theory calculations were done with the Gaussian program package [23], while the MOLPRO

program package [24] was used to perform all of the CASPT2 and QCISD(T) calculations.

2.2. High pressure kinetics

The high pressure capture kinetics for the barrierless R-1 and R-2 reactions was studied with the VRC-TST method, as implemented in the computer code VaReCoF [25]. These capture rate coefficients for R-1 and R-2 were also readily converted to high pressure dissociation rate coefficients via computations of the equilibrium constant. Detailed specifications of the VRC-TST approach can be found in [9–11]. Thus, only the details of the implementation for the present reactions are provided here.

In the VRC-TST approach, the dividing surfaces are defined in terms of a fixed separation between two pivot points, with one located on each fragment. For the R-1 reaction, the pivot points were displaced from the nitrogen atoms perpendicular to the plane of the NH_2 fragment and the C–N–H plane of the CH_3NH fragment. For the R-2 reaction, the pivot points were displaced from the carbon atoms perpendicular to the plane of the CH_3 fragment and displaced from the bonding nitrogen atom perpendicular to the H–N–N plane of the $NHNH_2$ fragment. In addition to these orbital centered pivot points employed for short separations of 4.5–8.5 au, this study also incorporated center-of-mass pivot points for large separations of 9–25 au.

For barrierless reactions, the form of the optimal dividing surface, as specified by the pivot point locations, may vary significantly as a function of total energy E, total angular momentum J, and temperature. Thus, both the pivot point locations and the separations between them were variationally optimized at each E and J to minimize the rate coefficient. The optimization of the separation R between the pivot points employed a 9-point-mesh with a constant grid spacing of 0.5 au for the short separations where the orbital centered pivot points are used. A separate 9-point-mesh with a variable grid spacing of 1.0–5.0 au was used for the large separations where the center-of-mass pivot points are used. For the orbital centered pivot points their displacement from the bonding atoms was variationally optimized on a 3×3 mesh with displacements of 0.5, 1.0, and 1.5 au for each of the pivot points.

As mentioned in Section 2.1, for the R-2 reaction there are two distinct addition sites on the $NHNH_2$ radical corresponding to the addition of the CH_3 radical to its front and back sides, and there are also front and back sides for the CH_3 radical. The front–front, front–back, back–front, and back–back capture rates are calculated separately, and the total capture rate is given by the sum of them. For the R-1 reaction, the two addition sites in CH_3NH are equivalent to one another because this radical is of Cs symmetry and the addition occurs out of the plane of symmetry. Similarly, the two addition sites on the NH_2 radical are equivalent. Consequently, the total capture rate is just four times the single-side capture rate.

For the R-1 reaction there is also a barrierless hydrogen abstraction channel, $NH_2 + CH_3NH \rightarrow NH_3 + CH_2NH$ where the NH_2 attacks the in-plane methyl hydrogen atom. In this study, we separate out the influence of this abstraction channel via the incorporation of an approximate infinite boundary potential. We also employ an approximate infinite boundary potential in separating out the front and back sides in R-2. The incorporation of such boundary potentials provides a satisfactory approximation to the fundamentally more meaningful multi-faceted dividing surface (MDS) approach [26].

Furthermore, the VRC-TST rate calculations were corrected for dynamical recrossing using the transmission coefficient of 0.85 obtained in a previous trajectory study of hydrocarbon radical–radical association reactions [13].

2.3. Pressure dependent kinetics

The pressure dependence of the MMH dissociation rate coefficients was determined by solving the two-dimensional master equation [27]. We employ N_2 as the bath gas since it is the major product in rocket engines fueled by MMH and thus should be present in high concentration. The energy transfer probability was approximated with a single-exponential-down model employing the temperature-dependent form $\Delta E_{down} = 200(T/300)^{0.85}\ cm^{-1}$ for the average downwards energy transfer. The approximation that ΔE_{down} increases roughly linearly with temperature has been validated in the previous study [28] for light bath gas molecules such as He, Ar, and N_2. The Lennard-Jones parameters for the MMH molecule are $\sigma = 4.4$ Å and $\varepsilon/k = 440\ cm^{-1}$, which are [7,8] based on the empirical method proposed by Wang and Frenklach [29]. For N_2, we used $\sigma = 3.62$ Å and $\varepsilon/k = 68\ cm^{-1}$ [30].

The energy levels for the underlying density and number of states calculations were evaluated on the basis of rigid-rotor harmonic oscillator assumptions for all but the torsional degrees of freedom. For the latter, hindered rotor corrections were obtained from one-dimensional fits to the torsional potentials employing Pitzer-Gwinn like approximations and the $I^{(2,3)}$ moments of inertia [31]. The fits to the torsional potentials were designed to reproduce the B3LYP/6-311++G(d,p) torsional frequency at the minimum as well as the QCISD(T)/CBS//B3LYP/6-311++G(d,p) torsional barrier heights and secondary minima. Three torsional degrees of freedom were considered here: the $CH_3 \cdots NHNH_2$

and $CH_3NH\cdots NH_2$ torsions in MMH, and the $CH_3\cdots NH$ torsion in CH_3NH. Meanwhile, the torsional frequency of 511 cm^{-1} in $NHNH_2$ was deemed large enough for a harmonic oscillator treatment to be accurate.

3. Results and discussion

3.1. Rate coefficients for R-1 and R-2

The VRC-TST capture rate coefficient for the $CH_3NH + NH_2$ association reaction is not sensitive to the basis set, showing less than 10% deviation between calculations with the cc-pVDZ and aug-cc-pVDZ basis sets. The combination of the aug-cc-pVTZ basis set correction and the geometry relaxation correction also have a minimal effect on the predictions, with the corrected results typically being a few percent greater, and a maximum increase of 10%.

The VRC-TST capture rate coefficient for the $CH_3 + NHNH_2$ association reaction, either for the front or back side, using the cc-pVDZ basis is 10–25% smaller than that using the aug-cc-pVDZ basis. Incorporating the one-dimensional aug-cc-pVTZ basis set correction, yields results that are within a few percent of those for the aug-cc-pVDZ, suggesting that the results for the latter basis are well converged. Meanwhile, the relaxation correction is more significant, increasing the rate coefficient by a factor ranging from 1.4 to 2.2 over the 500 to 2000 K temperature range. Prior studies [13] suggest that the overall predictions have an uncertainty on the order of 30% with roughly equal contributions arising from uncertainties in the underlying *ab initio* electronic structure calculations and from uncertainties in the VRC-TST assumptions.

As discussed in Section 2, for the $NH_2 + CH_3NH$ reaction, in addition to the addition reaction, there is a barrierless path leading to the hydrogen abstraction to yield $NH_3 + CH_2NH$. The incorporation of an assumed approximate infinite boundary potential, to exclude the orientations that would correlate with the hydrogen abstraction channel yields capture rate coefficients that are about 50% lower at 300 K. The influence of the abstraction channel decreases with increasing temperature and its exclusion reduces the predicted rate coefficient by less than 5% when the temperature is higher than 800 K. This decrease in importance is related to the fact that the transition state radical–radical separations decrease with increasing temperature. This small relative rate for the abstraction channel also suggests a similarly small contribution for the corresponding roaming channel in the reverse dissociation process.

The temperature dependence of the high pressure combination rate coefficients for R-1 and R-2 is shown in Fig. 3. The recombination rate coefficient for reaction R-1 is significantly greater than that for R-2 due to the differences in attractivity. The rate coefficient for R-1, denoted by k_{-1}^{∞}, is well fit by the expression $8.34 \times 10^{-10}\, T^{-0.429} \exp(20.1/T)$ cm^3 $molecule^{-1}$ s^{-1} over the temperature range of 200–2500 K. Similarly, the rate coefficient k_{-2}^{∞} for R-2 may be reproduced by the expression $3.99 \times 10^{-12}\, T^{0.085} \exp(404.2/T)$ cm^3 $molecule^{-1}$ s^{-1} over the same temperature range.

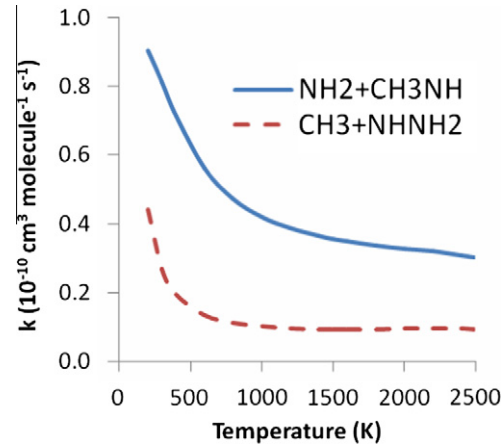

Fig. 3. Temperature dependence of the high pressure recombination rate coefficients for $NH_2 + CH_3NH$ (solid line) and $CH_3 + NHNH_2$ (dashed line).

3.2. MMH dissociation

The two simple bond fission channels R1 and R2 are expected to dominate the decomposition kinetics of MMH since the $CH_3NH + NH_2$ chan-

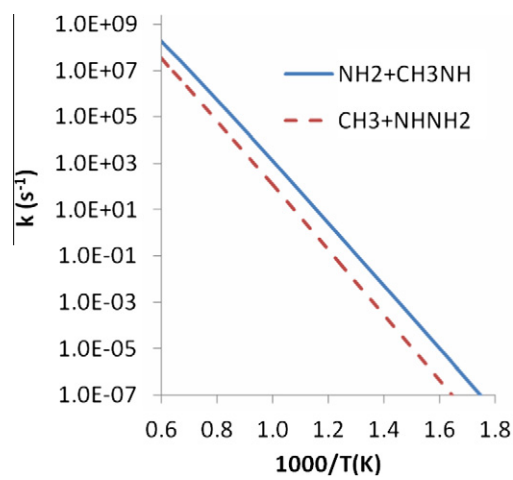

Fig. 4. Temperature dependence of the high pressure dissociation rate coefficients to produce $NH_2 + CH_3NH$ (solid line) or $CH_3 + NHNH_2$ (dashed line).

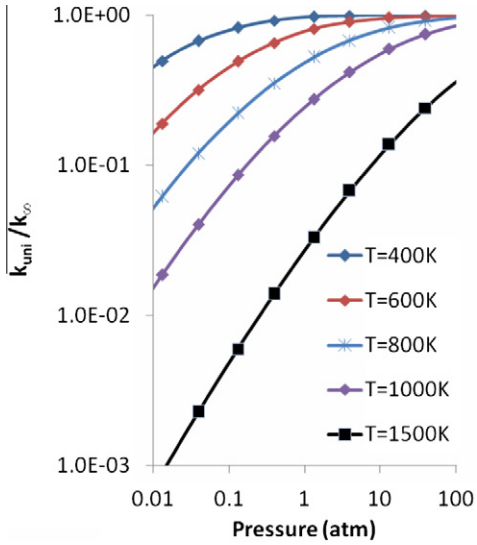

Fig. 5. Reduced falloff curves for MMH → $NH_2 + CH_3NH$ at different temperatures.

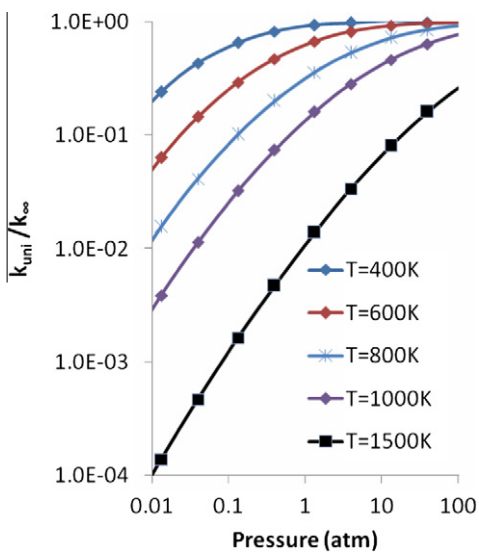

Fig. 6. Reduced falloff curves for MMH → $CH_3 + NHNH_2$ at different temperatures.

nel has the lowest threshold. Furthermore, the other three low lying channels have tight transition states [7]. The competitive dissociation to channels R1 and R2 from the MMH complex was modeled by solving the master equation and using the VRC transition state information discussed above.

The temperature dependence of the high pressure association rate coefficients for R1 and R2 is shown in Fig. 4. The falloff curves for these two reactions at different temperatures are shown in Figs. 5 and 6, respectively. The dominant product channel is R1 in the temperature range of 400–2500 K and the pressure range of 0.01–100 atm. The high- and low-pressure unimolecular rate coefficients for R1, denoted by k_1^∞ and k_1^0, respectively, are well reproduced by the expressions $4.55 \times 10^{23}\ T^{-2.147} \exp(-32560/T)\ s^{-1}$ and $1.17 \times 10^{36}\ T^{-11.89} \exp(-35987/T)\ cm^3\ molecule^{-1}\ s^{-1}$. The predicted falloff data for R1 can be fit to Troe's form [32,33] as $F_{cent} = 0.8671 \exp(-T/512) + 0.1329 \exp(-T/12283) + \exp(-7556/T)$, with the absolute maximum relative error and the least square error of 22% and 11%, respectively. Similarly, we have $k_2^\infty = 5.65 \times 10^{19}\ T^{-1.120} \exp(-33050/T)\ s^{-1}$, $k_2^0 = 2.36 \times 10^{29}\ T^{-10.37} \exp(-36161/T)\ cm^3\ molecule^{-1}\ s^{-1}$ and $F_{cent} = 0.404 \exp(-T/181205) + 0.596 \exp(-T/771) + \exp(-4390/T)$ to replicate the falloff data for R2 with a least square error of about 20%. Note that, for the higher of two dissociation channels, the low-pressure limit rate coefficient is not well defined. Instead, the quantity k_2^0 is simply an effective fitting parameter adjusted to produce an accurate fit to the falloff data over a given range of pressure, i.e., 1 Torr or greater.

The total MMH decomposition rate constant, obtained by summing over those to produce R1 and R2, is compared with previous experimental and simulation data in Fig. 7. Since Kerr et al. [3] measured the first-order rate coefficient for the homogeneous dissociation of the N–N bond of MMH, which is exactly reaction R1, their experimental data are appropriate for comparison of our theoretical results. Furthermore, most of their experimental data were measured at 0.02 atm, although the pressure range was reported as 0.01–0.04 atm. Figure 7 shows that

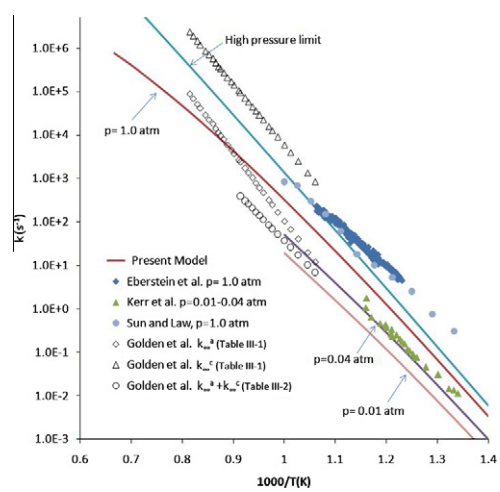

Fig. 7. Comparison of the predicted total decomposition rate constant through MMH → $NH_2 + CH_3NH$ and MMH → $CH_3 + NHNH_2$ with previous experimental and simulation data for the overall thermal decomposition of MMH.

our theoretical rate coefficients agree very well with the experimental data of Kerr et al., especially at 0.04 atm.

The very low pressure pyrolysis study of Golden et al. [5] measures the rate constant for unimolecular decomposition induced by wall collisions and then extracts high pressure rate constants via comparison with RRK and RRKM predictions. Their three different estimates for the high pressure rate constants, obtained for different assumptions in the RRKM calculations, span the range of the present predictions. Unfortunately, their directly measured unimolecular rate constants are not readily compared with the present pressure dependent predictions due to their convolution with gas-surface instead of gas-gas energy transfer effects.

To explain the disagreement between the present predictions and the previous experimental data of Eberstein and Glassman [4], we note that the latter are the total thermal decomposition rates of MMH in an adiabatic flow reactor at 750–1000 K and atmospheric pressure. In order to model these experiments, Sun and Law [7] developed an elementary reaction mechanism which consists of 43 species and 160 reactions. Although R1 and R2 are the most important reactions for MMH decomposition in their mechanism, the reactions $CH_3NHNH_2 + CH_3 \rightarrow CH_3NNH_2 + CH_4$ and $CH_3NNH_2 \rightarrow CH_3NNH + H$ have almost the same influence as R2, at least on the basis of sensitivity analysis. Consequently, the disagreement between the present predictions and the previous experimental and simulation data may be due to the absence of these qualitatively important reaction channels.

It is also perhaps worth noting that the agreement with experiment could be improved somewhat by employing larger energy transfer rates, but it is unlikely that a reasonable increase in that rate could increase the overall rates by more than a factor of two. Furthermore, it would have little if any effect at low temperatures and high pressures for the comparison with the data of Eberstein et al.

4. Conclusions

A theoretical analysis of the kinetics of the barrierless combination reactions $NH_2 + CH_3NH$ and $CH_3 + NHNH_2$, as well as the corresponding dissociation reactions of MMH (monomethylhydrazine), has been performed in the present study for a wide range of temperatures and pressures relevant to rocket engine combustion. Capture rate coefficients were determined by combining on-the-fly *ab initio* electronic structure calculations with variable reaction coordinate transition state theory. Master equation simulations were used to obtain the unimolecular reaction rates and branching ratios for MMH dissociation to the $NH_2 + CH_3NH$ and $CH_3 + NHNH_2$ channels, where the former channel is the dominant one.

Acknowledgements

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